Lesson 41: Discounted Reward and Bellman Equations

AI/ML Mathematics

Contents

1 Motivation and Overview 2

2 References 2

3 Discounted Return and Value Functions 3

4 The Bellman Expectation Operator 3

5 The Bellman Optimality Operator 4

6 Greedy Policies and Optimality 6

7 Policy Ordering and Improvement 6

8 Bellman Optimality Equations for Q^* 7

9 Exercises 9

10 Summary 9

11 Further Reading 10

1 Motivation and Overview

Consider a robot that must operate indefinitely in a warehouse, picking and delivering items day after day with no predetermined stopping time. Unlike the finite-horizon settings of Lesson 40, the agent faces an infinite sequence of decisions. How should it evaluate a policy that generates a never-ending stream of rewards? If the robot earns reward r every step forever, the total undiscounted reward is infinite and uninformative. The discounted reward criterion resolves this by weighting future rewards geometrically: a reward k steps in the future is worth γ^k times its face value, where $\gamma \in [0, 1)$.

This lesson develops the infinite-horizon discounted framework in full. We introduce the Bellman expectation and optimality operators, prove that these operators are contractions with unique fixed points, establish the existence and characterization of optimal policies, and derive the fundamental relationships between value functions and policies that underpin every RL algorithm.

Why the Discounted Bellman Framework Matters for RL/ML

- **Value iteration and policy iteration**: These classical algorithms (to be developed in a subsequent lesson) are iterative applications of the Bellman optimality and expectation operators, respectively.
- **Q-learning**: The Q-learning update $Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$ is a stochastic approximation to the Bellman optimality equation for Q^*.
- **Actor-critic methods**: The critic estimates V^π or Q^π by approximately solving the Bellman expectation equations; the actor improves the policy using the resulting value estimates.
- **Convergence guarantees**: The contraction property of Bellman operators is the key mathematical tool for proving convergence of both model-based and model-free RL algorithms.

Prerequisites. MDP model, policy classes, return, state-value function V^π, action-value function Q^π, advantage function, and the Bellman expectation equations (Lesson 39). Finite-horizon MDPs and backward induction (Lesson 40). Contraction mappings and the Banach fixed-point theorem on complete metric spaces (Lesson 4, Lesson 15). Norms on $\mathbb{R}^n$, in particular the supremum norm (Lesson 5).

Learning objectives. After completing this lesson, the student will be able to:

- *Define* the Bellman expectation operator $\mathcal{T}^\pi$ and the Bellman optimality operator $\mathcal{T}^*$.
- *Prove* that $\mathcal{T}^\pi$ and $\mathcal{T}^*$ are contraction mappings in the supremum norm.
- *Derive* that V^π is the unique fixed point of $\mathcal{T}^\pi$ and V^* is the unique fixed point of $\mathcal{T}^*$.
- *Prove* that the greedy policy with respect to V^* is optimal.
- *Establish* the policy ordering theorem and its consequences.
- *Compute* V^* and Q^* for small MDPs by solving the Bellman optimality equations.

2 References

- Puterman [2014, Chapters 5–6] – Definitive treatment of the discounted infinite-horizon criterion, Bellman operators, and optimality.

- Bertsekas [2012, Chapter 1] – Dynamic programming operators, contraction properties, and convergence of iterative methods.
- Blackwell [1962] – Foundational paper establishing the contraction property and existence of optimal stationary policies.
- Denardo [1967] – Unified contraction framework for dynamic programming operators.
- Sutton and Barto [2018, Chapters 3–4] – Bellman equations from the RL perspective with intuitive explanations.
- Howard [1960] – Policy iteration and its connection to the Bellman expectation equations.

3 Discounted Return and Value Functions

We work throughout with a finite MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with $|\mathcal{S}| = S$, $|\mathcal{A}| = A$, $R_{\max} = \max_{s,a} |R(s,a)|$, and $\gamma \in [0,1)$. Value functions are elements of $\mathbb{R}^S$, which we equip with the supremum norm $\|v\|_\infty = \max_{s\in\mathcal{S}} |v(s)|$.

Recall from Lesson 39 (Definition 5.1 and Proposition 5.1 therein) that the discounted return $G_t = \sum_{k=0}^{\infty} \gamma^k R(S_{t+k}, A_{t+k})$ is well-defined and bounded by $R_{\max}/(1-\gamma)$, and that the state-value function $V^\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$ and action-value function $Q^\pi(s,a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$ satisfy the Bellman expectation equations.

We now introduce the operator-theoretic viewpoint.

4 The Bellman Expectation Operator

Definition 4.1 (Bellman Expectation Operator)**.** For a stationary policy π, the *Bellman expectation operator* $\mathcal{T}^\pi : \mathbb{R}^S \to \mathbb{R}^S$ is defined by

$$(\mathcal{T}^\pi v)(s) = \sum_{a\in\mathcal{A}} \pi(a \mid s) \left[R(s,a) + \gamma \sum_{s'\in\mathcal{S}} P(s' \mid s,a)\, v(s') \right]$$

for all $s \in \mathcal{S}$ and $v \in \mathbb{R}^S$.

Remark 4.2. In matrix notation, $\mathcal{T}^\pi v = r^\pi + \gamma P^\pi v$, where $r^\pi \in \mathbb{R}^S$ has entries $r^\pi_s = \sum_a \pi(a \mid s) R(s,a)$ and P^π is the $S \times S$ transition matrix with entries $P^\pi_{ss'} = \sum_a \pi(a \mid s) P(s' \mid s,a)$.

Theorem 4.3 ($\mathcal{T}^\pi$ Is a Contraction)**.** *For any stationary policy π, the operator $\mathcal{T}^\pi$ is a contraction mapping on $(\mathbb{R}^S, \|\cdot\|_\infty)$ with modulus γ. That is, for all $u, v \in \mathbb{R}^S$,*

$$\|\mathcal{T}^\pi u - \mathcal{T}^\pi v\|_\infty \le \gamma \|u - v\|_\infty.$$

Proof. For any state s,

$$\begin{aligned}
|(\mathcal{T}^\pi u)(s) - (\mathcal{T}^\pi v)(s)| &= \left| \sum_a \pi(a \mid s) \gamma \sum_{s'} P(s' \mid s,a)[u(s') - v(s')] \right| \\
&\le \gamma \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s,a)\, |u(s') - v(s')| \\
&\le \gamma \sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s,a)\, \|u - v\|_\infty \\
&= \gamma \|u - v\|_\infty \underbrace{\sum_a \pi(a \mid s) \sum_{s'} P(s' \mid s,a)}_{=1}.
\end{aligned}$$

Taking the maximum over s gives $\|\mathcal{T}^\pi u - \mathcal{T}^\pi v\|_\infty \le \gamma \|u - v\|_\infty$. □

Theorem 4.4 (V^π Is the Unique Fixed Point of $\mathcal{T}^\pi$)**.** *For any stationary policy π, the value function V^π is the unique fixed point of $\mathcal{T}^\pi$ in $\mathbb{R}^S$. That is, $\mathcal{T}^\pi V^\pi = V^\pi$, and if $\mathcal{T}^\pi v = v$ for some $v \in \mathbb{R}^S$, then $v = V^\pi$.*

Proof. **Fixed point property:** The Bellman expectation equation from Lesson 39 (Theorem 6.1 therein) states exactly that $V^\pi(s) = \sum_a \pi(a \mid s)[R(s,a)+\gamma \sum_{s'} P(s' \mid s,a)V^\pi(s')]$ for all s, which is $\mathcal{T}^\pi V^\pi = V^\pi$.

Uniqueness: Since $(\mathbb{R}^S, \|\cdot\|_\infty)$ is a complete metric space (it is a finite-dimensional normed space) and $\mathcal{T}^\pi$ is a contraction with modulus $\gamma < 1$ (Theorem 4.3), the Banach fixed-point theorem guarantees that $\mathcal{T}^\pi$ has a unique fixed point. □

Characterization of V^π via Fixed Point

The value function V^π is completely characterized as the unique solution to the equation $v = \mathcal{T}^\pi v$, equivalently $v = r^\pi + \gamma P^\pi v$, equivalently $v = (I - \gamma P^\pi)^{-1} r^\pi$. This provides three methods for computing V^π: iterating the operator (successive approximation), solving the linear system, or matrix inversion.

Example 4.5 (Policy Evaluation via Iteration)**.** Consider the two-state MDP from Lesson 39 (Example 6.3 therein) with $\gamma = 0.5$ under policy $\pi(s_1) = a_2$, $\pi(s_2) = a_2$. Then $r^\pi = (10, 2)^\top$ and $P^\pi = \left(\begin{smallmatrix} 0 & 1 \\ 0.3 & 0.7 \end{smallmatrix}\right)$. Starting from $v^{(0)} = (0, 0)^\top$:

$$\begin{aligned}
v^{(1)} &= r^\pi + 0.5 P^\pi v^{(0)} = (10, 2)^\top, \\
v^{(2)} &= (10, 2)^\top + 0.5 \begin{pmatrix} 0 & 1 \\ 0.3 & 0.7 \end{pmatrix} \begin{pmatrix} 10 \\ 2 \end{pmatrix} = (10, 2)^\top + (1, 2.2)^\top = (11, 4.2)^\top, \\
v^{(3)} &= (10, 2)^\top + 0.5 \begin{pmatrix} 4.2 \\ 6.24 \end{pmatrix} = (12.1, 5.12)^\top.
\end{aligned}$$

The iterates converge to $V^\pi = (I - 0.5P^\pi)^{-1} r^\pi = (14, 6)^\top$, consistent with Lesson 39.

5 The Bellman Optimality Operator

Definition 5.1 (Bellman Optimality Operator)**.** The *Bellman optimality operator* $\mathcal{T}^* : \mathbb{R}^S \to \mathbb{R}^S$ is defined by

$$(\mathcal{T}^* v)(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a)\, v(s') \right]$$

for all $s \in \mathcal{S}$ and $v \in \mathbb{R}^S$.

Lemma 5.2 (Max-Norm Inequality for Max Operator)**.** *For any functions $f, g : \mathcal{A} \to \mathbb{R}$ on a finite set $\mathcal{A}$, $|\max_a f(a) - \max_a g(a)| \le \max_a |f(a) - g(a)|$.*

Proof. Let $a^* = \arg\max_a f(a)$. Then $\max_a f(a) - \max_a g(a) = f(a^*) - \max_a g(a) \le f(a^*) - g(a^*) \le |f(a^*) - g(a^*)| \le \max_a |f(a) - g(a)|$. By symmetry, $\max_a g(a) - \max_a f(a) \le \max_a |f(a) - g(a)|$. Combining gives the result. □

Theorem 5.3 ($\mathcal{T}^*$ Is a Contraction)**.** *The Bellman optimality operator $\mathcal{T}^*$ is a contraction mapping on $(\mathbb{R}^S, \|\cdot\|_\infty)$ with modulus γ:*

$$\|\mathcal{T}^* u - \mathcal{T}^* v\|_\infty \le \gamma \|u - v\|_\infty \quad \textit{for all } u, v \in \mathbb{R}^S.$$

Proof. For any state s, define $f_s(a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)u(s')$ and $g_s(a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)v(s')$. Then $(\mathcal{T}^* u)(s) = \max_a f_s(a)$ and $(\mathcal{T}^* v)(s) = \max_a g_s(a)$.

By Lemma 5.2,

$$\begin{aligned}
|(\mathcal{T}^* u)(s) - (\mathcal{T}^* v)(s)| &\le \max_a |f_s(a) - g_s(a)| \\
&= \max_a \left| \gamma \sum_{s'} P(s' \mid s,a)[u(s') - v(s')] \right| \\
&\le \gamma \max_a \sum_{s'} P(s' \mid s,a) \|u - v\|_\infty \\
&= \gamma \|u - v\|_\infty.
\end{aligned}$$

Taking the maximum over s yields the result. □

Theorem 5.4 (Bellman Optimality Equations Characterize V^*)**.** *The optimal value function V^* is the unique fixed point of $\mathcal{T}^*$:*

$$V^*(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^*(s') \right] \quad \textit{for all } s \in \mathcal{S},$$

and no other $v \in \mathbb{R}^S$ satisfies $\mathcal{T}^* v = v$.

Proof. We establish the result in three steps.

Step 1: $\mathcal{T}^*$ has a unique fixed point. Since $(\mathbb{R}^S, \|\cdot\|_\infty)$ is complete and $\mathcal{T}^*$ is a contraction (Theorem 5.3), the Banach fixed-point theorem guarantees a unique fixed point $v^* \in \mathbb{R}^S$.

Step 2: $V^* \le v^*$. For any stationary deterministic policy d, the Bellman expectation equation gives $V^d = \mathcal{T}^d V^d$. Since $(\mathcal{T}^d v)(s) \le (\mathcal{T}^* v)(s)$ for all v and s (the max is at least as large as any particular action), we have $\mathcal{T}^d \le \mathcal{T}^*$ pointwise. Now consider the iterates $v^{(n)} = (\mathcal{T}^*)^n v^{(0)}$ starting from $v^{(0)} = V^d$:

$$V^d = \mathcal{T}^d V^d \le \mathcal{T}^* V^d = v^{(1)}.$$

By monotonicity of $\mathcal{T}^*$ (if $u \le v$ componentwise then $\mathcal{T}^* u \le \mathcal{T}^* v$), we get $v^{(1)} \le v^{(2)} \le \cdots$, and since $v^{(n)} \to v^*$, we conclude $V^d \le v^*$ for all d. Taking the supremum over d gives $V^* \le v^*$.

Step 3: $v^* \le V^*$. Let d^* be the greedy policy with respect to v^*: $d^*(s) = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)v^*(s')]$. Since $v^* = \mathcal{T}^* v^*$, we have $v^*(s) = R(s,d^*(s)) + \gamma \sum_{s'} P(s' \mid s,d^*(s))v^*(s') = (\mathcal{T}^{d^*} v^*)(s)$, so v^* is a fixed point of $\mathcal{T}^{d^*}$. By Theorem 4.4, the unique fixed point of $\mathcal{T}^{d^*}$ is V^{d^*}, so $v^* = V^{d^*} \le V^*$.

Combining Steps 2 and 3: $V^* = v^*$. □

Bellman Optimality Equation

The optimal value function is the unique solution to:

$$V^*(s) = \max_{a \in \mathcal{A}} \left[R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a)\, V^*(s') \right], \quad s \in \mathcal{S}.$$

Equivalently, the optimal action-value function satisfies:

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q^*(s',a'), \quad (s,a) \in \mathcal{S} \times \mathcal{A}.$$

These are nonlinear fixed-point equations (due to the max) and generally cannot be solved in closed form, motivating iterative algorithms such as value iteration.

Bellman Optimality Equations Are Nonlinear

Unlike the Bellman expectation equations for V^π, which form a *linear* system $v = r^\pi + \gamma P^\pi v$ solvable by matrix inversion, the Bellman optimality equations are *nonlinear* due to the max operator. Direct solution is generally intractable for large state spaces. This is why iterative methods (value iteration, policy iteration) and approximate methods (function approximation, deep RL) are essential.

6 Greedy Policies and Optimality

Definition 6.1 (Greedy Policy)**.** Given a value function $v \in \mathbb{R}^S$, the *greedy policy* with respect to v is the deterministic policy d_v defined by

$$d_v(s) = \arg\max_{a\in\mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, v(s') \right].$$

Similarly, given an action-value function $q \in \mathbb{R}^{S\times A}$, the greedy policy with respect to q is $d_q(s) = \arg\max_a q(s,a)$.

Theorem 6.2 (The Greedy Policy with Respect to V^* Is Optimal)**.** *Let π^* be the greedy policy with respect to V^*. Then π^* is optimal: $V^{\pi^*}(s) = V^*(s)$ for all $s \in \mathcal{S}$. Equivalently, the greedy policy with respect to Q^* satisfies $\pi^*(s) = \arg\max_a Q^*(s,a)$.*

Proof. This was established in Step 3 of the proof of Theorem 5.4: the greedy policy d^* with respect to V^* satisfies $\mathcal{T}^{d^*}V^* = \mathcal{T}^*V^* = V^*$, so V^* is the fixed point of $\mathcal{T}^{d^*}$, hence $V^{d^*} = V^*$.

For the Q^* formulation: since $Q^*(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)V^*(s')$, we have $\arg\max_a Q^*(s,a) = \arg\max_a [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)V^*(s')] = d_{V^*}(s) = \pi^*(s)$. □

Greedy Policies in RL Algorithms

Theorem 6.2 is the theoretical foundation for value-based RL methods. Q-learning aims to approximate Q^*; once a good approximation $\hat{Q} \approx Q^*$ is obtained, the agent acts greedily: $a = \arg\max_a \hat{Q}(s,a)$. DQN, Double DQN, and Dueling DQN all implement this strategy with neural network function approximation. The theorem guarantees that greedy action selection is optimal when $\hat{Q} = Q^*$ exactly.

7 Policy Ordering and Improvement

Definition 7.1 (Policy Ordering)**.** For two stationary policies π and π', we write $\pi \geq \pi'$ if $V^\pi(s) \geq V^{\pi'}(s)$ for all $s \in \mathcal{S}$.

Proposition 7.2 (Policy Ordering Is a Partial Order)**.** *The relation $\geq$ on stationary policies is a partial order: reflexive ($\pi \geq \pi$), antisymmetric ($\pi \geq \pi'$ and $\pi' \geq \pi$ imply $V^\pi = V^{\pi'}$), and transitive ($\pi \geq \pi'$ and $\pi' \geq \pi''$ imply $\pi \geq \pi''$).*

Proof. Reflexivity and transitivity follow directly from the corresponding properties of $\geq$ on $\mathbb{R}$. For antisymmetry: if $V^\pi(s) \geq V^{\pi'}(s)$ and $V^{\pi'}(s) \geq V^\pi(s)$ for all s, then $V^\pi = V^{\pi'}$ as functions. □

Theorem 7.3 (Policy Improvement Theorem)**.** *Let π be a stationary policy and let π' be the greedy policy with respect to V^π:*

$$\pi'(s) = \arg\max_{a\in\mathcal{A}} Q^\pi(s,a) = \arg\max_{a\in\mathcal{A}} \left[R(s,a) + \gamma \sum_{s'} P(s' \mid s,a)\, V^\pi(s') \right].$$

Then $\pi' \geq \pi$, i.e., $V^{\pi'}(s) \geq V^{\pi}(s)$ for all $s \in \mathcal{S}$. Moreover, equality $V^{\pi'} = V^{\pi}$ holds if and only if $V^{\pi} = V^*$.

Proof. For any state s,

$$\begin{aligned} V^{\pi}(s) &= \sum_a \pi(a \mid s) Q^{\pi}(s,a) \leq \max_a Q^{\pi}(s,a) = Q^{\pi}(s,\pi'(s)) \\ &= R(s,\pi'(s)) + \gamma \sum_{s'} P(s' \mid s,\pi'(s))\, V^{\pi}(s'). \end{aligned}$$

Define $\delta(s) = (\mathcal{T}^{\pi'} V^{\pi})(s) - V^{\pi}(s) \geq 0$ for all s. Then $V^{\pi} = \mathcal{T}^{\pi'} V^{\pi} - \delta$ where $\delta \geq 0$. Iterating the operator $\mathcal{T}^{\pi'}$:

$$\begin{aligned} V^{\pi} &\leq \mathcal{T}^{\pi'} V^{\pi} \\ &\leq (\mathcal{T}^{\pi'})^2 V^{\pi} \\ &\leq \cdots \\ &\leq \lim_{n\to\infty} (\mathcal{T}^{\pi'})^n V^{\pi} = V^{\pi'}, \end{aligned}$$

where each inequality uses the monotonicity of $\mathcal{T}^{\pi'}$ (if $u \leq v$ then $\mathcal{T}^{\pi'} u \leq \mathcal{T}^{\pi'} v$) and the limit equals $V^{\pi'}$ by the Banach fixed-point theorem.

For the "moreover" part: $V^{\pi'} = V^{\pi}$ implies $V^{\pi}(s) = \max_a Q^{\pi}(s,a)$ for all s, which is the Bellman optimality equation, hence $V^{\pi} = V^*$. Conversely, if $V^{\pi} = V^*$, then π is already optimal and $V^{\pi'} = V^* = V^{\pi}$. □

Policy Improvement in Policy Iteration

The policy improvement theorem is the engine of the policy iteration algorithm: starting from any policy π_0, compute V^{π_0} (policy evaluation), form the greedy policy π_1 (policy improvement), and repeat. Since $V^{\pi_{k+1}} \geq V^{\pi_k}$ and there are finitely many deterministic policies, the sequence must converge to V^* in at most A^S steps. In practice, convergence is much faster.

8 Bellman Optimality Equations for Q^*

Definition 8.1 (Bellman Optimality Operator for Q)**.** The *Bellman optimality operator for action-values* $\mathcal{T}_Q^* : \mathbb{R}^{S\times A} \to \mathbb{R}^{S\times A}$ is

$$(\mathcal{T}_Q^* q)(s,a) = R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} q(s',a').$$

Proposition 8.2 ($\mathcal{T}_Q^*$ Is a Contraction with Fixed Point Q^*)**.** *The operator $\mathcal{T}_Q^*$ is a contraction on $(\mathbb{R}^{S\times A}, \|\cdot\|_\infty)$ with modulus γ, and Q^* is its unique fixed point.*

Proof. For any $q, q' \in \mathbb{R}^{S\times A}$ and any (s,a),

$$\begin{aligned} |(\mathcal{T}_Q^* q)(s,a) - (\mathcal{T}_Q^* q')(s,a)| &= \gamma \left| \sum_{s'} P(s' \mid s,a) \left[\max_{a'} q(s',a') - \max_{a'} q'(s',a') \right] \right| \\ &\leq \gamma \sum_{s'} P(s' \mid s,a) \max_{a'} |q(s',a') - q'(s',a')| \\ &\leq \gamma \|q - q'\|_\infty. \end{aligned}$$

The first inequality uses Lemma 5.2 and the triangle inequality. Taking the max over (s, a) gives $\|\mathcal{T}_Q^* q - \mathcal{T}_Q^* q'\|_\infty \le \gamma \|q - q'\|_\infty$. By the Banach fixed-point theorem, $\mathcal{T}_Q^*$ has a unique fixed point.

To show this fixed point is Q^*: from Theorem 5.4, $V^*(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^*(s')]$. Setting $Q^*(s, a) = R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^*(s')$, we verify that $(\mathcal{T}_Q^* Q^*)(s, a) = R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) \max_{a'} Q^*(s', a') = R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^*(s') = Q^*(s, a)$. □

Example 8.3 (Solving the Bellman Optimality Equation for a Two-State MDP)**.** Revisit the two-state MDP from Lesson 39 with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{a_1, a_2\}$, $\gamma = 0.5$:

s	a	$R(s, a)$	$P(s_1 \mid s, a)$	$P(s_2 \mid s, a)$
s_1	a_1	5	0.5	0.5
s_1	a_2	10	0	1
s_2	a_1	-1	0.8	0.2
s_2	a_2	2	0.3	0.7

The Bellman optimality equations are:

$$V^*(s_1) = \max\{5 + 0.5[0.5V^*(s_1) + 0.5V^*(s_2)],\ 10 + 0.5[0 \cdot V^*(s_1) + 1 \cdot V^*(s_2)]\},$$
$$V^*(s_2) = \max\{-1 + 0.5[0.8V^*(s_1) + 0.2V^*(s_2)],\ 2 + 0.5[0.3V^*(s_1) + 0.7V^*(s_2)]\}.$$

Guess that $\pi^*(s_1) = a_2$ and $\pi^*(s_2) = a_2$ (the policy found optimal in Lesson 39). Then the equations become:

$$V^*(s_1) = 10 + 0.5V^*(s_2),$$
$$V^*(s_2) = 2 + 0.5[0.3V^*(s_1) + 0.7V^*(s_2)].$$

From the second: $V^*(s_2)(1 - 0.35) = 2 + 0.15V^*(s_1)$, so $0.65V^*(s_2) = 2 + 0.15V^*(s_1)$. Substituting $V^*(s_1) = 10 + 0.5V^*(s_2)$: $0.65V^*(s_2) = 2 + 0.15(10 + 0.5V^*(s_2)) = 3.5 + 0.075V^*(s_2)$, giving $0.575V^*(s_2) = 3.5$, so $V^*(s_2) = 6.087 \approx 6.09$. Then $V^*(s_1) = 10 + 0.5 \cdot 6.087 = 13.043 \approx 13.04$.

We verify the guess: at s_1, action a_1 gives $5 + 0.5(0.5 \cdot 13.04 + 0.5 \cdot 6.09) = 5 + 4.78 = 9.78 < 13.04$, confirming a_2 is optimal. At s_2, action a_1 gives $-1 + 0.5(0.8 \cdot 13.04 + 0.2 \cdot 6.09) = -1 + 5.82 = 4.82 < 6.09$, confirming a_2 is optimal.

Example 8.4 (Gambling Problem)**.** A gambler has wealth $s \in \mathcal{S} = \{0, 1, \ldots, 100\}$ and at each step wagers $a \in \{0, 1, \ldots, \min(s, 100 - s)\}$. With probability $p = 0.4$ wealth increases by a; with probability 0.6 it decreases by a. States 0 and 100 are absorbing. Rewards are $R(s, a) = 0$ except $R(100, \cdot) = 1$. The Bellman optimality equation for $s \in \{1, \ldots, 99\}$ is $V^*(s) = \max_a [pV^*(s+a) + (1-p)V^*(s-a)]$, with $V^*(0) = 0$, $V^*(100) = 1$. Solving by value iteration reveals the optimal policy stakes aggressively at wealth levels that are powers of 2.

We now consolidate the structural results of this lesson.

Fundamental Theorem of MDPs

For any finite MDP with $\gamma \in [0, 1)$: (i) there exists a unique V^*, the fixed point of $\mathcal{T}^*$; (ii) there exists a deterministic stationary optimal policy π^*; (iii) π^* is the greedy policy w.r.t. V^* (or Q^*); (iv) no history-dependent or randomized policy improves upon π^*. This follows from the contraction property (Theorem 5.3), the fixed-point characterization (Theorem 5.4), and the greedy optimality result (Theorem 6.2).

9 Exercises

Exercise 9.1 (Bellman Operator Properties). $(*)$
Prove that the Bellman optimality operator $\mathcal{T}^*$ is monotone: if $u(s) \leq v(s)$ for all $s \in \mathcal{S}$, then $(\mathcal{T}^* u)(s) \leq (\mathcal{T}^* v)(s)$ for all s.

Exercise 9.2 (Value Iteration Convergence). $(*)$
Let $v^{(0)} = 0 \in \mathbb{R}^S$ and $v^{(n+1)} = \mathcal{T}^* v^{(n)}$ for $n \geq 0$.

(a) Show that $\|v^{(n)} - V^*\|_\infty \leq \gamma^n R_{\max}/(1-\gamma)$.

(b) How many iterations are needed to guarantee $\|v^{(n)} - V^*\|_\infty \leq \varepsilon$?

Exercise 9.3 (Bellman Equation for Stochastic Policies). $(**)$
Let π be a stationary randomized policy.

(a) Prove that $\mathcal{T}^\pi$ is still a γ-contraction even when π is stochastic.

(b) Show that $(\mathcal{T}^* v)(s) \geq (\mathcal{T}^\pi v)(s)$ for all $v \in \mathbb{R}^S$, $s \in \mathcal{S}$, and any stochastic π.

(c) Conclude that $V^*(s) \geq V^\pi(s)$ for all stochastic policies π.

Exercise 9.4 (Solving Bellman Equations for a Three-State MDP). $(**)$
Consider the MDP with $\mathcal{S} = \{1, 2, 3\}$, $\mathcal{A} = \{L, R\}$, $\gamma = 0.9$:

s	a	$R(s,a)$	$P(1 \mid s,a)$	$P(2 \mid s,a)$	$P(3 \mid s,a)$
1	L	0	0.5	0.5	0
1	R	1	0	0	1
2	L	2	0.3	0.4	0.3
2	R	0	0	0	1
3	L	0	0	0	1
3	R	0	0	0	1

(a) Evaluate the policy $\pi(1) = R$, $\pi(2) = L$, $\pi(3) = L$ by solving the linear Bellman equations.

(b) Write down the Bellman optimality equations and solve them for V^*.

(c) Determine the optimal policy and verify by checking all $2^3 = 8$ deterministic policies.

Exercise 9.5 (Policy Improvement Proof Variant). $(***)$
Prove the following strengthened version of Theorem 7.3: if π is a stochastic stationary policy that is not optimal, then there exists a state $\bar{s}$ such that $V^{\pi'}(\bar{s}) > V^\pi(\bar{s})$, where π' is the greedy deterministic policy. That is, strict improvement occurs at some state whenever π is suboptimal.

Hint: If $V^{\pi'} = V^\pi$ then V^π satisfies the Bellman optimality equation, contradicting $\pi \neq \pi^*$.

10 Summary

- The *Bellman expectation operator* $\mathcal{T}^\pi v = r^\pi + \gamma P^\pi v$ is a γ-contraction in the sup-norm; its unique fixed point is V^π.
- The *Bellman optimality operator* $(\mathcal{T}^* v)(s) = \max_a [R(s,a) + \gamma \sum_{s'} P(s' \mid s,a) v(s')]$ is also a γ-contraction; its unique fixed point is V^*.
- Both contraction properties rely on $\gamma < 1$ and the stochasticity of P; the Banach fixed-point theorem then guarantees existence, uniqueness, and iterative computability.

- The greedy policy with respect to V^* (or Q^*) is optimal (Theorem 6.2), justifying value-based RL methods.
- The *policy improvement theorem*: greedifying w.r.t. V^π yields a policy at least as good, with strict improvement unless π is already optimal.
- Every finite discounted MDP admits a unique V^* and at least one deterministic stationary optimal policy.
- The Bellman optimality equations are nonlinear (due to max) and motivate iterative algorithms: value iteration, policy iteration, and their approximate variants.

11 Further Reading

- **Lesson on TD Learning** shows how to solve Bellman equations without a model, using sample-based updates.
- **Lesson on Policy Gradients** develops an alternative that directly optimizes parameterized policies.
- Puterman [2014, Chapters 6–7] develops value iteration and policy iteration with convergence rates and stopping criteria.
- Denardo [1967] provides a unified contraction framework for discounted, weighted, and other reward criteria.
- Blackwell [1962] proves that for large enough γ, a single policy is optimal for all discount factors above a threshold (*Blackwell optimality*).
- Bertsekas [2012, Chapters 8–9] extends Bellman operators to approximate DP with function approximation – the theoretical basis of deep RL.

References

Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control*, volume II. Athena Scientific, 4th edition, 2012.

David Blackwell. Discrete dynamic programming. *The Annals of Mathematical Statistics*, 33 (2):719–726, 1962.

Eric V. Denardo. Contraction mappings in the theory underlying dynamic programming. *SIAM Review*, 9(2):165–177, 1967.

Ronald A. Howard. Dynamic programming and markov processes. *MIT Press*, 1960.

Martin L. Puterman. *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons, 2nd edition, 2014.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.